清华大学热能工程系 2005 级“燃烧学”期末考试试卷

考试时间： 2008 年 6 月 17 日 班级：_____

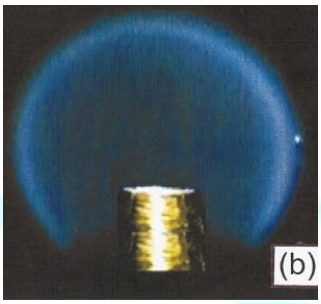
学号：_____ 姓名：_____

Instructions(提示):

1. 考试为开卷考试，允许带教科书，作业，课件，笔记本和另外的参考书各一样，但不能互借，答题必须独立完成。（The examination is **OPEN BOOKS**. Calculators, textbook, homework and notebook are allowed, but they can not be lent to others. You should answer the questions independently and help is not allowed）
2. 试卷共有5题，你要在120分钟内完成这次考试，先检查一下你的卷子，题目是否对，共有3页。答案必须写在答题卷上，试题卷上回答无效。答题必须明确，清晰，计算题应给出单位。（There are five questions on this exam. You have up to 120 minutes to work this exam. Please be sure that you have a complete exam package. Please work exam problems on the pages provided. Show your work neatly and write your answers. Give units for all answers requiring them.）
3. 如果试卷不对，请立即与监考教师说明。（If a page is missing, take the exam to a proctor immediately.）
4. 强烈推荐你先读一下所有的问题，并对回答问题的时间进行安排。答卷完成后请再仔细检查一下。（Budget your time for each question and be sure to first answer those questions that you know how to work. Check your work after completing the exam.）
5. 请你现在就在答卷纸和试卷纸上写上你的**名字与学号**，并在你的答卷纸上标明你的试卷的类型（A或B卷）（**PRINT** your name **NOW** in the spaces at the top of your pages.）
6. 答卷可以用中文或英文，如果是双语的，以中文为准。试卷有英译的，也以中文为准。(The answers can be in English or Chinese. If two languages are used in one answer, I will take Chinese as the selection. Some of the problems have been translated in English. If there are any difference between Chinese and English, Chinese is for your choice.)

1　分析与回答题（Answer the following questions briefly）**（每题5分， 8题，共40分）**

1.1　对于层流非预混火焰(采用本生灯)，通常情况下都是狭长形射流形状，当出现下图所示火焰结构时，你能判断产生这种火焰的原因吗？并请判断此时流场的 Reynolds 数和 Froude 数处在什么条件下？As for laminar non-premixed flame (Bunsen burner), the jet flame often behaves a narrow and long shape. However, as you see the flame shape shown in the figure below, can you judge its reason? Moreover, what is status of Reynolds number and Froude number in this case?



1.2　碳燃烧过程中，经常采用 $\mathbf{R}_C(kg/s-m^2)=k_C MW_C[O_{2,s}]$ 这个总包反应来描述碳反应速率，结合表面化学反应机理，请分析这种反应速率和氧气浓度(或分压)呈线性比例关系成立的条件是什么？ As far as carbon burning is concerned, the global reaction of $\mathbf{R}_C(kg/s-m^2)=k_C MW_C[O_{2,s}]$ is often used to describe the carbon burning rate. On the basis of the mechanism of surface reaction, please give the reason that supports the linear relationship between carbon burning rate and oxygen concentration (or oxygen partial pressure)!

1.3　一个本生灯预混火焰，在实验中一定的流速下出现了推举现象，此时将一个十字交叉的钢丝放在喷口出口处，试分析会出现什么现象，为什么？In the experiment of Bunsen premixed flame, the liftoff occurs at a relatively high flow velocity. If we place an X-shaped crossing steel-wire at the exit of Bunsen burner, please analyze what will be occurred then! And why?

1.4　碳烟通常产生于非预混火焰，请列出柴油、汽油、乙烯和甲烷这四种燃料在非预混火焰中产生碳烟的概率排序？请结合碳烟形成机理给出扼要的分析。Soot usually produces from the non-premixed flames. Please list the soot forming tendency of ethylene, diesel, gasoline and methane and briefly give your reason.

1.5　一个释热 1000kW 的射流燃烧器，火焰长度为 L，根据工业要求，如果需将火焰长度减小 1/4 而释热率保持 1000kW 不变的话，则如何改进燃烧器设计，并给出您的原因。The jet flame with heat release 1000kW has a length, L. If the flame length is changed to 1/4 of L to meet the industrial needs and the heat release rate is kept constant, how do you modify the design of the burner? Give your reason.

1.6　北京早期没有天然气，家中燃用人工煤气，主要成分为 $CO+H_2$，后来改用天然气（主

要成分为甲烷，环境可以得到改善，试分析原来燃用人工煤气的家用炉具是否可以直接燃用天然气？为什么 ？The man-made coal gas, mainly containing CO and H_2, is used for the coking during an earlier period of non-supply of natural-gas at Beijing. And then the natural gas replace the coal gas in order to improve environmental quality. Do you think the cooktop burner of coal gas can be directly used to burn the natural gas? Give you reason!

1.7　其他条件相同时，一颗燃料液滴在空气中燃烧和在氧气（体积分数为 21%）与二氧化碳的混合物中燃烧，哪个速度快？（Determine which burnout rate is faster. One droplet is burned in air and another is burned in the mixture of Oxygen and CO_2 (mole ratio 21% :79%). The other condition is same.）

1.8　一种纯碳颗粒(Car=100%)，一种是仅由灰分和碳成分组成的、干燥的煤颗粒(Aar=20%; Car=80%)，假设燃烧过程中煤颗粒中灰分和碳分密度相同且均匀分布，而且高温下随着燃烧进展煤颗粒随着质量的损失而等密度地减小体积(即缩核现象)。在扩散控制条件下且传递数、扩散系数和碳颗粒温度都接近，试比较同样初始粒径大小的纯碳颗粒和煤颗粒的燃烧寿命不同。There are two kinds of coal particles. One is pure carbon (Car=100%), and the other is dry mixture of carbon and ash (i.e., Aar=20%; Car=80%). It is assumed that the ash content and carbon content of latter coal sample are equably distributed, which enable this kind of coal burn following a law that the particle volume reduction is linear to the mass loss *without density variation* (this is known as shrinking-core model). **Please compare the lifetime of two coals under a condition of identical initial-diameter**! *Assuming the diffusion-controlled regime, the similar transfer number B, mass diffusivity D and surface temperature* T_s.

2　(12分) 对于圆口燃烧器，Roper(1977)提出非预混火焰长度的经验公式，即 $L_f = 1330\dfrac{Q_F(T_\infty/T_F)}{\ln(1+1/S)}$，式中 Q_F 为燃料体积流率(m^3/s)；环境温度 T_∞、燃料温度 T_F 都为300K；1大气压。如果给定质量流率满足 $\dot{m}_F = 4\times10^{-6}$ kg/s，丙烷、甲烷、氮气和空气的摩尔质量分别为44.10、16.04、28.01和28.85g/mol。试求：

Roper (1977) proposed an empirical equation, $L_f = 1330\dfrac{Q_F(T_\infty/T_F)}{\ln(1+1/S)}$, to predict the non-premixed flame length of circular ports. In the equation, Q_F is volumetric flowrate. The ambience and fuel temperature, T_∞ and T_F, are kept at 300K; pressure is 1 atm. If mass flowrate, $\dot{m}_F$, is given as 4×10^{-6} kg/s, please try to obtain:

(a) 60%丙烷和40%甲烷组成混和燃料气在空气下的火焰长度；

the flame length for a mixed fuel gas composing of 60% propane and 40% methane;

(b) 60%丙烷被40%氮气稀释后的混和气在空气下的火焰长度；

the flame length for a mixed fuel gas composing of 60% propane and 40% Nitrogen;

(c) 60%丙烷和40%空气组成的混合气在空气下的火焰长度；

the flame length for a mixed fuel gas composing of 60% propane and 40% air;

3 （24分）假设一个液滴在低于液滴沸点的温度下在高温烟气中蒸发，有热量传导到液滴内部，建议可用“洋葱皮”模型计算加热过程的影响。（环境温度为T_∞）.Assuming a droplet evaporating at a temperature that is lower than its boiling point. If the heat is conducted into the droplet interior, the *onion-skin model* can be used to evaluate this kind of effect.

3.1 （9分）试证明其质量蒸发速率为：

Please derive the mass evaporation rate as equation below:

$$\dot{m}_F = \frac{4\pi k_g r_s}{c_{pg}} \ln\left[1 + \frac{c_{pg}(T_\infty - T_s)}{q_{i-l} + h_{fg}}\right]$$

3.2 （8分）由此计算一个直径500微米的正辛烷（C_8H_{18}）液滴，在常压（1atm）高温烟气中蒸发，试估计其质量蒸发速率和液滴寿命。且初始液滴的整体温度是300K，环境烟气温度为1800K。高温烟气的物性可以用空气来表示，正辛烷液体的比热为4kJ/kg-K。C_{pg}=1200J/kg-K, k_g=0.1W/m-K。

Try to calculate both **evaporation rate** and **lifetime** of a 500μm-diameter n-octane droplet (C_8H_{18}) at 1atm, high temperature flue-gas ambience! The initial temperature of droplet is 300K, and the temperature of flue gas is 1800K. The properties of flue gas can be used by those of air; the specific heat of liquid droplet is 4kJ/kg-K, and the gas specific heat C_{pg} is 1200J/kg-K; the gas thermal conductivity is k_g=0.1W/m-K.

3.3 （7分）比较其他条件相同的条件下，计算如果不考虑热量传导到液滴内部的情形的质量蒸发速率和液滴寿命（忽略内部加热，假设液滴初始温度为液体的沸点）。

If we do not consider the conduction into the droplet interior (that is, the initial temperature of droplet is assumed as the boiling pint temperature), please calculate the both **evaporation rate** and **lifetime** of the droplet! The rest conditions during droplet evaporation are same.

4 (13分) 直径250μm的碳颗粒在静止空气($Y_{O_2,\infty} = 0.233$)中燃烧，燃烧模式由扩散阻力控制，条件：压力为101325Pa；碳表面温度T_s=1500K；$k_c = 3\times10^5 \exp(-17966/T_s)$ [m/s]；$D(293K) = 1.6\cdot10^{-5}$ [m^2/s]。下面计算均基于单膜模型来开展。

A 250μm-diameter carbon particle is burned at the quiescent air ($Y_{O_2,\infty} = 0.233$) following a diffusion-controlled regime. The conditions are: pressure at 101325Pa; surface temperature at T_s=1500K; $k_c = 3\times10^5 \exp(-17966/T_s)$ [m/s]; $D(293K) = 1.6\cdot10^{-5}$ [m^2/s]. And the single-film model is used here.

(a) 如果气相混合物的摩尔质量假设为MW_{mix}=32 kg/kmol，请计算出碳燃烧速率$\dot{m}_C$(5分)；

If the molecular weight of gas mixture is MW_{mix}=32 kg/kmol, please determine the carbon burning rate $\dot{m}_C$ (5分);

(b) 试计算扩散控制条件下，碳表面处的CO_2的质量分数$Y_{CO_2,s}$，提示：从CO_2的扩散方程出发建单膜模型并和O_2的对比(5分);
Under the diffusion-controlled regime, please determine the mass fraction of CO_2 at carbon surface, $Y_{CO_2,s}$. (5分). Clue: this work can be started by derive the diffusion equation of CO_2 in the single-film model and compare it with that of O_2.

(c) 富氧循环燃烧条件下(环境气体O_2/CO_2=25/75)，仍然是扩散阻力控制条件下，试求$Y_{CO_2,s}$。
If now the carbon is burned in the ambience of the mixed O_2/CO_2=25/75 (oxy-fuel combustion), please determine the $Y_{CO_2,s}$ again.

5 （11分）实验中用一维火焰的方法测量H_2-空气预混火焰燃烧的速度，一维火焰在一个圆管内实现，预混气体流过管内，管的面积不变，测量时火焰的位置不动，假设管子为绝热。混合物的初温为298K，当量比Φ=0.8，常压下测量，此时测得燃烧后烟气的流出速度为10m/s，试计算火焰的传播速度。

The method of one-dimensional flat flame is attempted to measure the flame speed of H_2-air flame. This method is performed by introduce H_2-air unburned mixture to pass a cylindrical tube with constant cross-sectional area. The position of flame front is fixed during the measurement and the tube is adiabatic. The initial temperature of the mixture is 298K (with equivalence Φ =0.8 and 1atm). At this time, if the effluent flue gas velocity is measured 10m/s, please determine the flame speed, S_L.